

LAKATRASK, Sweden

WMO: 021740

Lat: 66.2789N

Lon: 21.1285E

Elev: 607

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.9	-19.4	-31.7	0.9	-24.6	-26.1	1.3	-19.3	12.4	27.6	11.3	26.9	0.2	40	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	77.7	61.8	74.1	60.0	70.7	58.7	64.7	73.7	62.5	70.0	60.4	67.6	5.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	83.4	67.3	59.4	77.2	65.6	57.3	71.6	63.4	30.0	73.4	28.4	70.1	27.0	67.3	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.2	9.1	-28.5	81.5	5.2	4.8	-32.2	84.9	-35.3	87.7	-38.2	90.3	-42.0	93.8		
(5)			-29.2	66.8	5.0	2.7	-32.8	68.8	-35.7	70.3	-38.5	71.9	-42.2	73.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	8.7	10.1	20.2	32.5	43.5	52.4	58.4	54.4	45.2	32.2	20.4	13.9		
	DBStd	19.73	14.12	13.95	10.31	7.02	6.91	6.44	5.42	6.06	6.38	8.67	11.60	13.77		
	HDD50	6874	1281	1116	924	525	225	44	3	25	171	552	888	1120		
	HDD65	11775	1746	1536	1389	975	666	381	214	333	595	1017	1338	1585		
	CDD50	589	0	0	0	0	24	117	263	160	25	0	0	0		
	CDD65	16	0	0	0	0	0	3	10	3	0	0	0	0		
	CDH74	328	0	0	0	0	15	80	184	49	0	0	0	0		
	CDH80	46	0	0	0	0	1	16	27	2	0	0	0	0		
	WSAvg	3.6	2.5	2.9	3.9	4.3	4.7	4.5	4.0	3.7	3.9	3.3	2.7	2.8		
(15) Precipitation	PrecAvg	22.8	1.5	1.3	1.0	1.1	1.6	2.5	3.2	2.6	2.3	1.9	2.1	1.5		
	PrecMax	32.0	2.7	2.2	1.9	2.4	3.1	4.4	4.6	4.8	8.4	3.4	3.7	3.2		
	PrecMin	15.7	0.2	0.4	0.4	0.5	0.4	0.2	1.4	0.9	0.7	0.9	0.9	0.5		
	PrecStd	4.4	0.6	0.5	0.5	0.5	0.7	1.1	0.9	1.1	1.8	0.8	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	43.0	47.7	60.5	76.0	81.8	83.2	79.0	66.4	53.8	45.1	40.5		
		MCWB	37.5	38.0	39.2	46.3	56.7	62.3	65.0	66.0	56.2	47.2	41.7	36.7		
	2%	DB	34.7	37.9	42.9	52.9	70.3	75.1	78.7	74.5	62.2	50.0	41.5	36.3		
		MCWB	32.7	33.9	36.0	42.4	55.0	58.0	63.1	61.5	54.5	45.8	39.4	34.0		
	5%	DB	31.7	33.6	39.8	49.2	63.7	70.7	75.3	70.8	59.3	47.2	37.7	33.9		
		MCWB	30.8	31.4	34.3	40.2	51.1	56.4	61.7	60.3	52.8	44.4	36.5	32.6		
	10%	DB	28.2	30.5	36.3	45.7	58.5	66.6	71.5	66.8	56.5	44.2	34.9	32.2		
		MCWB	27.4	29.2	31.9	38.2	47.5	54.5	60.2	58.4	51.5	41.9	33.9	31.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	38.5	39.6	46.8	59.0	63.9	67.7	67.6	58.1	49.0	42.3	37.3		
		MCDB	41.0	42.9	46.7	57.8	72.2	76.1	77.4	75.1	62.8	51.4	43.8	40.5		
	2%	WB	33.1	34.0	36.8	43.4	55.8	61.0	65.4	64.1	56.0	46.8	39.2	34.6		
		MCDB	34.1	36.8	41.7	51.7	68.5	71.2	75.0	71.5	60.4	49.4	41.0	35.9		
	5%	WB	30.8	31.6	34.7	40.8	52.0	58.1	63.5	61.6	54.4	44.5	36.4	32.8		
		MCDB	31.8	33.4	39.1	48.4	63.5	67.1	71.7	67.8	57.7	46.8	37.5	33.8		
	10%	WB	27.4	29.1	32.4	38.5	48.5	55.6	61.5	59.3	52.7	42.1	34.0	31.3		
		MCDB	28.2	30.8	35.7	45.1	56.8	64.0	68.7	65.2	55.6	44.1	34.8	32.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	18.5	21.2	18.6	20.3	19.8	19.8	19.4	17.3	13.5	12.6	14.7		
		MCDBR	12.9	16.7	20.4	24.1	31.1	28.1	26.7	25.3	22.3	16.1	11.7	12.1		
		MCWBR	12.1	14.3	14.9	14.5	17.0	13.3	13.0	13.9	14.9	12.2	10.4	11.4		
	5% WB	MCDBR	13.0	15.5	19.6	22.7	29.4	23.5	22.3	21.0	17.4	14.0	11.1	12.2		
		MCWBR	12.3	13.6	14.8	13.9	16.5	12.2	12.5	12.6	12.4	11.1	10.1	11.6		
(40) Clear-Sky Solar Irradiance	taub		0.188	0.257	0.290	0.322	0.336	0.339	0.350	0.334	0.304	0.282	0.338	N/A		
	taud		1.948	2.154	2.230	2.308	2.469	2.487	2.470	2.501	2.564	2.403	2.720	N/A		
	Ebn at Noon		77	191	245	267	276	277	269	262	247	187	82	N/A		
	Edh at Noon		5	16	25	30	29	30	29	25	19	13	5	N/A		
(44) All-Sky Solar Radiation	RadAvg		27	207	691	1286	1573	1658	1542	1142	644	236	49	3		
	RadStd		2	30	52	63	158	130	149	103	94	40	7	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page